
IT SIMPLERA INSTITUTE

Cyber security Analyst Internship program



INTERNSHIP WEEKLY REPORT

SUBMITTED BY	MALIHA GUL
ROLE	RED TEAM INTERN
INSTITUTE	IT-SIMPLERA INSTITUTE
INSTRUCTOR	TALAL REHAN
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QUESTION NO:1

Network Information

Screenshot

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26100.1]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-JG3NK4T
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Ethernet Connection (10) I219-LM
    Physical Address. . . . . : 74-78-27-3F-A3-4D
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : E0-D4-64-E7-4A-4C
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:
```

```
Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
    Physical Address. . . . . : E2-D4-64-E7-4A-4B
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
    Physical Address. . . . . : E0-D4-64-E7-4A-4B
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::86a9:b4bc:e0da:498c%17(Preferred)
    IPv4 Address. . . . . :
    Subnet Mask . . . . . :
    Lease Obtained. . . . . : Friday, July 10, 2026 2:22:10 PM
    Lease Expires . . . . . : Monday, July 13, 2026 2:22:09 PM
    Default Gateway . . . . . :
    DHCP Server . . . . . :
    DHCPv6 IAID . . . . . : 232838244
    DHCPv6 Client DUID. . . . . : 00-01-00-01-30-AB-B9-6D-E0-D4-64-E7-4A-4B
    DNS Servers . . . . . :
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\DELL>
```

IPv4 Address

An IPv4 address is a unique address assigned to a device on a network. It allows the device to communicate with other devices over the Internet or a local network.

AC Address (Physical Address)

A MAC address is a unique hardware address assigned to a network adapter. It identifies the device on a local network.

Default Gateway

A Default Gateway is the router that forwards data from the local network to other networks, such as the Internet.

QUESTION:2

Basic Networking Commands

1. Ipconfig /all

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26100.1]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-JG3NK4T
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Ethernet Connection (10) I219-LM
    Physical Address. . . . . : 74-78-27-3F-A3-4D
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : E0-D4-64-E7-4A-4C
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:
```

```

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . . . . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : E2-D4-64-E7-4A-4B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . . . . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : E0-D4-64-E7-4A-4B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::86a9:b4bc:e0da:498c%17(Preferred)
IPv4 Address. . . . . : 
Subnet Mask . . . . . : 
Lease Obtained. . . . . : Friday, July 10, 2026 2:22:10 PM
Lease Expires . . . . . : Monday, July 13, 2026 2:22:09 PM
Default Gateway . . . . . : 

DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 232838244
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-AB-B9-6D-E0-D4-64-E7-4A-4B
DNS Servers . . . . . : 
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\DELL>

```

Purpose:

Displays detailed network configuration, including IP address, MAC address, DNS servers, and Default Gateway.

2. Ping google.com

```

C:\Users\DELL> ping google.com

Pinging google.com [142.250.203.174] with 32 bytes of data:
Reply from 142.250.203.174: bytes=32 time=71ms TTL=115
Reply from 142.250.203.174: bytes=32 time=62ms TTL=115
Reply from 142.250.203.174: bytes=32 time=68ms TTL=115
Reply from 142.250.203.174: bytes=32 time=58ms TTL=115

Ping statistics for 142.250.203.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 58ms, Maximum = 71ms, Average = 64ms

```

Purpose:

Tests whether the computer can communicate with Google's server and measures the response time.

3. tracert google.com

```
C:\Users\DELL> tracert google.com

Tracing route to google.com [142.250.203.174]
over a maximum of 30 hops:

  1    5 ms    1 ms    1 ms  192.168.1.1 [192.168.1.1]
  2   14 ms   13 ms   14 ms  172.20.1.1 [172.20.1.1]
  3   16 ms   11 ms   11 ms  10.12.12.1 [10.12.12.1]
  4   34 ms   35 ms   37 ms  172.16.235.113 [172.16.235.113]
  5   31 ms   27 ms   25 ms  59.103.180.253
  6   16 ms   18 ms   20 ms  182.176.222.45
  7   18 ms   20 ms   18 ms  10.253.12.26 [10.253.12.26]
  8   37 ms   37 ms   36 ms  10.253.4.18 [10.253.4.18]
  9   39 ms   38 ms   39 ms  10.253.4.8 [10.253.4.8]
 10   53 ms   52 ms   52 ms  72.14.212.112
 11   57 ms   55 ms   60 ms  172.253.51.55
 12   68 ms   79 ms   79 ms  142.251.50.215
 13   83 ms   74 ms   81 ms  pnfjra-ah-in-f14.1e100.net [142.250.203.174]

Trace complete.
```

Purpose:

Shows the path (hops) that data packets take from the computer to Google's server.

4. nslookup google.com

```
C:\Users\DELL> nslookup google.com
Server: 192.168.1.1
Address: 192.168.1.1

Non-authoritative answer:
Name: google.com
Addresses: 2a00:1450:4019:817::200e
          142.250.203.174
```

Purpose:

Retrieves the IP address of a domain name using the Domain Name System (DNS).

5. arp -a

```
C:\Users\DELL> arp -a

Interface: 192.168.1.8 --- 0x11
    Internet Address      Physical Address      Type
    192.168.1.1           0c-37-47-83-17-53     dynamic
    192.168.1.255         ff-ff-ff-ff-ff-ff     static
    224.0.0.22            01-00-5e-00-00-16     static
    224.0.0.251           01-00-5e-00-00-fb     static
    224.0.0.252           01-00-5e-00-00-fc     static
    239.255.255.250       01-00-5e-7f-ff-fa     static
    255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Purpose:

Displays the ARP table, showing the mapping of IP addresses to MAC addresses on the local network.

QUESTION:3

Router vs. Switch

Feature	Router	Switch
Function	Connects different networks	Connects devices within the same network
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer)
Address Used	IP Address	MAC Address
Data Forwarding	Routes packets between networks	Forwards frames within a LAN
Internet Access	Provides Internet connectivity	Does not provide Internet access by itself
Common Use	Home and office networks	Connecting computers, printers, and other devices in a LAN

QUESTION: 4

IPv4 vs. IPv6

Feature	IPv4	IPv6
Address Length	32-bit	128-bit
Address Format	Decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:db8::1)
Number of Addresses	About 4.3 billion	Almost unlimited
Configuration	Manual or DHCP	Auto-configuration supported
Security	Optional IPsec	Built-in IPsec support
Header	Simpler but less efficient	Simpler and more efficient

QUESTION:5

OSI Model

Layer	Primary Function	Common Protocol	Networking Device
1. Physical	Transmits raw bits through cables	Ethernet	Hub
2. Data Link	Transfers frames using MAC addresses	Ethernet	Switch
3. Network	Routes packets between networks	IP	Router
4. Transport	Ensures reliable data transfer	TCP	Firewall
5. Session	Establishes and manages communication sessions	NetBIOS	Gateway

6. Presentation	Data encryption and formatting	SSL/TLS	Gateway
7. Application	Provides network services to applications	HTTP	PC